SQL ASSIGNMENT 10

Subject: Database Management Systems (SQL) **Topic:** Subqueries & Nested Queries **Status:** Completed

QUESTION 1

Write a SQL query using a subquery in the `WHERE` clause to find all restaurants whose rating is higher than the average rating of all restaurants in the city.

```
SELECT name, city, rating FROM Restaurants r1 WHERE rating > ( SELECT AVG(rating) FROM Restaurants r2 WHERE r2.city = r1.city );
```

QUESTION 2

Write a SQL query using a subquery in the `SELECT` statement to display each user's name alongside their total number of orders placed (Zomato user profile summary).

```
SELECT u.name AS user_name, ( SELECT COUNT(*) FROM Orders o WHERE o.user_id = u.id ) AS total_orders FROM Users u;
```

QUESTION 3

Given a `Movies` table and a `Reviews` table, write a SQL query using `IN` with a subquery to list all movies that have at least one 5-star review (BookMyShow top-rated section).

```
SELECT title, release_year FROM Movies WHERE id IN ( SELECT movie_id FROM Reviews WHERE rating = 5 );
```

QUESTION 4

Write a nested SQL query to find the names of sellers who have sold products in every category listed in a `Categories` table (Relational Division).

```
-- Universal Quantification: Find sellers where NO category exists that they haven't sold in SELECT s.name AS seller_name FROM Sellers s WHERE NOT EXISTS ( SELECT c.id FROM Categories c WHERE NOT EXISTS ( SELECT p.id FROM Products p WHERE p.seller_id = s.id AND p.category_id = c.id ) );
```